Benedict Armstrong: Curriculum Vitae

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Abstract

Computational Science and Engineering student at ETH Zürich working at the intersection of machine learning and physics. My research focuses on sequence models for detecting and characterising gravitational-wave signals. I also have several years of experience shipping full-stack software as a freelancer and have taught algorithms, data structures and software engineering to hundreds of students.

1 Introduction

Born and raised in Vienna, currently living in Zürich. In my free time, I like to code, tinker with electronics, design and make furniture, and generally enjoy all types of DIY/maker projects. I am also interested in FPV drones, IoT prototyping, 3D printing, tennis, cycling, triathlons, and of course, all things computer science.

Our contributions are as follows:

- **Research.** Machine learning for gravitational-wave physics with Computational Applied Mathematics & AI Lab (CAMAIL) and the LIGO Lab at MIT (Section 2), presented at CERN [2].
- **Publications.** First author of PhyTS, a benchmark for scientific time series at NeurIPS 2026 [1].
- **Engineering.** Freelance full-stack software for clients in Zürich (Section 3) and independent products (Section 5).
- **Teaching.** Four teaching roles across ETH Zürich and TU Wien (Section 4).

2 Research: AI for Gravitational-Wave Physics

Setting. ML Research Intern at the Computational Applied Mathematics & AI Lab (CAMAIL), ELLIS Institute Tübingen and Max Planck Institute for Intelligent Systems, in collaboration with the LIGO Lab at MIT and supervised by T. Konstantin Rusch (June 2025 – Summer 2026).

Explainer. When two neutron stars spiral into each other, they send ripples through spacetime that LIGO measures as length changes far smaller than a proton. These signals can last minutes and are buried in non-stationary, glitchy detector noise, while astronomers need an alert within seconds to point telescopes at the merger. I work on Linear Oscillatory State-Space Models (LinOSS): sequence models that handle very long inputs efficiently and make it possible to detect binary neutron star mergers in real time.

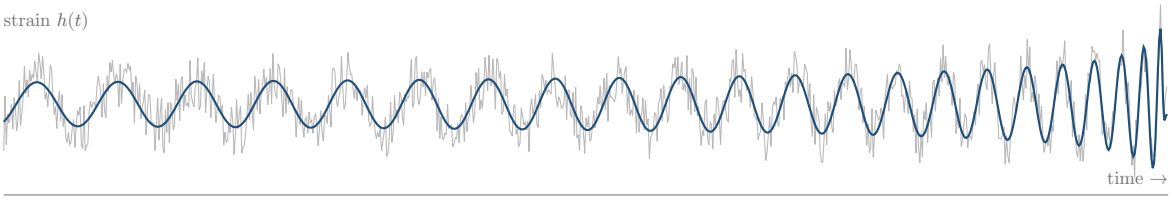


Figure 1: Schematic binary neutron star inspiral: the signal's frequency and amplitude grow as the stars approach merger (blue), while the detector records it buried in noise (grey). Illustrative, not real data.

Contributions.

- Designed and trained state-space models for low-latency binary neutron star detection on LIGO strain data.
- Built the training and evaluation pipeline in PyTorch (Lightning) and JAX/Equinox.
- Co-developed PhyTS, a benchmark of physics time-series tasks (NeurIPS 2026).

Talks. The detection work was presented at the AI for Gravitational Waves Workshop at CERN [2]; my colleague Kevin Yoon presented our state-space approach to parameter estimation at the Fast Machine Learning for Science Conference 2026 [3].

Tools. Python, PyTorch, PyTorch Lightning, JAX, Equinox.

3 Experience

Freelance Software Engineer, Armstrong Systems (2022 – 2024). Founded Armstrong Systems, software consulting for clients in Zürich. Full-stack development of various projects including a web-based data management tool, visualization dashboards and large-scale data ingestion pipelines.

- Scoped, built and maintained a web-based data management tool used day to day by the client's team.
- Built interactive visualization dashboards on top of the clients' operational data.
- Designed large-scale data ingestion pipelines, from source integration to deployment and monitoring.

Coding Assistant for LehrFEM++ Finite Element Library, Seminar for Applied Mathematics, ETH Zürich (Fall 2024). Coding assistant for the C++ finite element library LehrFEM++, used in the course *Numerical Methods for Partial Differential Equations* taught by Prof. Dr. Ralf Hiptmair.

- Extended and maintained the LehrFEM++ codebase used by students in the course's programming exercises.
- Wrote documentation and examples to make the library easier for students to pick up.

Paramedic (civil service), Samariterbund (2018 – 2019). Worked as a paramedic in Vienna as part of my civil service.

Various summer jobs, Various (Summers 2017–2021). Various summer jobs in Vienna, including working as a postman, food delivery driver and math/CS tutor for high school students.

4 Teaching

All teaching roles are summarised in Table 1. As head teaching assistant for Computer Science II, I coordinated the team of teaching assistants and the exercise classes and prepared exercise material with the lecturers. At TU Wien, I co-developed the Capture the Flag (CTF) challenges for *Security for Systems Engineering* and supervised students during the competition.

Table 1: Teaching positions, grouped by institution.

Role	Course	Period
Department of Computer Science, ETH Zürich		
Teaching Assistant	Software Engineering Fundamentals	Spring Semester 2024
Head Teaching Assistant	Computer Science II	August 2023 – February 2024
Teaching Assistant	Computer Science II	August 2022 – February 2023
Research Group Industrial Software (INSO), TU Wien		
Teaching Assistant and Full Stack Developer	Security for Systems Engineering	Spring 2021

5 Projects

Presio (presio.ch). Web app for presenting PDF slides with a presenter controller and a mirrored audience window. PDFs stay in the browser by default; signed-in users can share a session code so others can follow live on their own devices.

AcceptOdds (acceptodds.com). Short description of what AcceptOdds does, who it is for and what is technically interesting about it.

6 Education

Table 2: Degrees.

Degree	Programme	Institution	Period
M.Sc.	Computational Science and Engineering (Robotics)	ETH Zürich	2026 – present
B.Sc.	Computational Science and Engineering (Robotics)	ETH Zürich	2021 – 2025
B.Sc.	Computer Science (Software & Information Engineering)	Vienna University of Technology	2019 – 2021 (switch to ETH)

7 Skills

Languages. English, German.

Technologies. Python, PyTorch, JAX, C++, C, Java, JavaScript, PHP, SQL, Angular, React, NextJS, Svelte(Kit), FastAPI, REST, HTML, CSS, Mongo, Postgres, Docker, Git, Arduino/ESP8266/32 based hardware development, Scrum.

Acknowledgements

Research supervised by T. Konstantin Rusch, in collaboration with the LIGO Lab at MIT. Thanks to all co-authors of PhyTS [1].

References

[1] **Benedict Armstrong**, Jeroen Audenaert, Hannah P. Binney, Alice Cheng, Maarten De Vos, Luis F. Domingues, Allison Eto, Mary Feliz, Joseph Formaggio, Jessica Fry, Philip Harris, Ilay Kamai, Erik Katsavounidis, Mykyta Kliapets, Konstantinos Kontras, Aobo Li, Paul Liang, Daniel Muthukrishna, Christina Reissel, Daniela Rus, T. Konstantin Rusch, Avi Shporer, Penny L. Slocum, Talia E. Weiss, Lindley Winslow, Kyungseop Yoon. PhyTS: A Benchmark for Scientific Time Series. In *Advances in Neural Information Processing Systems (NeurIPS)*, Sydney, 2026.

[2] Benedict Armstrong. Linear Oscillatory State-Space Models for Binary Neutron Star Detection. Talk at *AI for Gravitational Waves Workshop*, CERN, Geneva, May 2026.

[3] Kevin (Kyungseop) Yoon. Direct Parameter Estimation of Binary Neutron Star Mergers with State Space Models. Talk at *Fast Machine Learning for Science Conference 2026*, UC San Diego, September 2026.